The documentation and process conversion measures necessary to comply with this revision shall be completed by 12 February 1994

INCH-POUND

MIL-S-19500/395C
12 November 1993
SUPERSEDING
MIL-S-19500/395B
1 February 1984

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPES 2N3735, 2N3735L, and 2N3737, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon switching transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (similar to TO-5 and TO-46).

1.3 Maximum ratings.

TYPE	P_T T_A = +25°C	P_T T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	$R_{\theta JC}$	T_{OP} and T_{STG}
	W	W	V dc	V dc	V dc	A dc	°C/mW	°C
2N3735	1.0 1/	2.9 3/	75	40	5	1.5	.060	
2N3737	0.5 2/	1.9 4/	75	40	5	1.5	.088	-55 to +200

1/ Derate linearly 5.71 mW/°C above T_A = +25°C.
2/ Derate linearly 2.86 mW/°C above T_A = +25°C.
3/ Derate linearly 16.6 mW/°C above T_C = +25°C.
4/ Derate linearly 11.3 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics.

	h_{FE3} 1/	$\|h_{fe}\|$	$V_{CE(sat)}$	Cobo	Pulse response		
LIMITS	V_{CE} = 1.0 V dc I_C = 0.5 A dc	V_{CE} = 10 V dc I_C = 50 mA dc f = 100 MHz	I_C = 500 mA dc I_B = 50 mA dc	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	t_r	t_d	t_{off}
			V dc	pF	ns	ns	ns
MIN	40	2.5					
MAX	140	6.0	0.5	9.0	8.0	40	60

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, Appendix F, Figure 8 (TO-5), T-1C for 2N3735 and T-1A for 2N3735L, and Figure 9 (TO-46), T-2B for 2N3737.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening. Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CBO1} and h_{FE3}	Not applicable
11	I_{CBO1} and h_{FE3}; ΔI_{CBO1} = 100% of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = 15% of initial value.	I_{CBO1} and h_{FE3}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100% of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = ±15% of initial value.	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = ±15% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5 (all product assurance levels); JANS level, V_{CB} = 10 V dc; 2N3735 and 2N3735L P_T = 1.0 W, 2N3737 - P_T = 0.5 W. JANTX and JANTXV levels V_{CB} = 30 V dc; 2N3735 and 2N3735L P_T = 1.0 W, 2N3737 - P_T = 0.5 W.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in IVa (JANS) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
B3	2037	Test condition A.
B4	1037	V_{CB} = 10 V dc; P_T = 1.0 W for 2N3735, 2N3735L, P_T = 0.5 W for 2N3737 at T_A = +30°C ± 5°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 V dc; P_T = 1.0 W, for 2N3735 and 2N3735L, P_T = 0.5 W for 2N3737 at T_A = +125°C ± 25°C or adjusted T_A to give an average lot T_J = 275°C.

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4.4.3 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
B3	1027	V_{CB} = 30 V dc; P_T = 1.0 W for 2N3735 and 2N3735L, P_T = 0.5 W for 2N3737 at T_A = +30°C ± 5°C. No heat sink or forced-air cooling on the devices shall be permitted.
B3	2037	Test condition A.

4.4.4 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	V_{CB} = 30 V dc, P_T = 1.0 W for 2N3735 and 2N3735L, P_T = 0.5 W for 2N3737 at T_A = +30°C ± 5°C. No heat sink or forced-air cooling on device shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance (to be performed for qualification only). Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 45 mA dc for 2N3735 and 2N3735L and 31 mA dc for 2N3737.

b. Collector-emitter voltage shall magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C.

e. Mounting arrangements shall be with heat sink to case.

f. Maximum limit shall be 60°C/watt for 2N3735, 2N3735L, and 88°C/watt for 2N3737.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	40		V dc
Breakdown voltage collector to base	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$	75		V dc
Breakdown voltage emitter to base	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	5		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO1}		250	nA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 30 V dc; V_{EB} = 2.0 V dc	I_{CEX1}		200	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4.0 V dc	I_{EBO1}		100	nA dc
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc	h_{FE1}	35		
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE2}	40		
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE3}	40	140	
Forward-current transfer ratio	3076	V_{CE} = 1.5 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE4}	20	80	
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.5 A dc; pulsed (see 4.5.1)	h_{FE5}	20		
Collector to emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{CE(sat)1}$		0.2	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.3	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		0.5	V dc
Collector to emitter voltage (saturated)	3071	I_C = 1.0 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)4}$		0.9	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)1}$		0.8	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.0	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)3}$		1.2	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 1.0 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)4}$	0.9	1.4	V dc
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 30 V dc; V_{EB} = 2.0 V dc	I_{CEX2}		250	µA dc
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE6}	15		

See footnote at end of table.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Magnitude of common - emitter small-signal short- circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 100 MHz	$\|h_{fe}\|$	2.5	6.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		9.0	pF
Input capacitance (output open - circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		80	pF
Pulse response:						
Delay time	3251	Test condition A; V_{CC} = 30 V dc; V_{BE} = 2 V dc; I_C = 1.0 A dc; I_{B1} = 100 mA dc; (see figure 1)	t_d		8.0	ns
Rise time	3251	Test condition A; V_{CC} = 30 V dc; V_{BE} = 2 V dc; I_C = 1.0 A dc; I_{B1} = 100 mA dc; (see figure 1)	t_r		40	ns
Turn-off time	3251	Test condition A; V_{CC} = 30 V dc; I_C = 1.0 A dc; I_{B1} = $-I_{B2}$ = 100 mA dc; (see figure 2)	t_{off}		60	ns
Subgroups 5, 6, and 7						
Not applicable						

For sampling plan see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements.

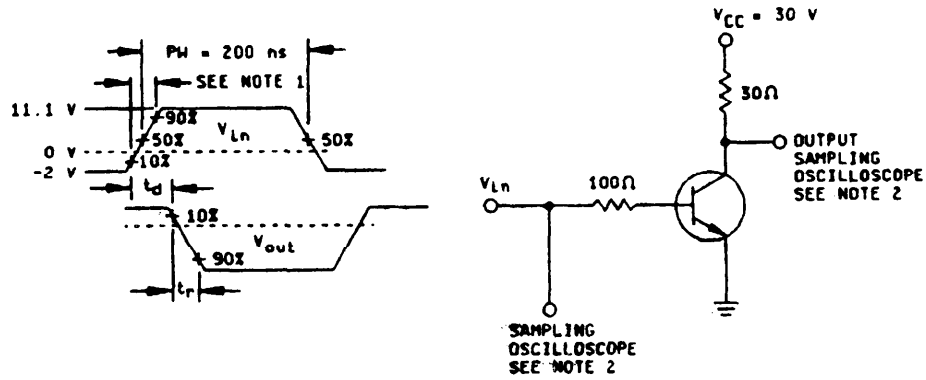
Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO1}		250	nA dc
2.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO1}		300	nA dc
3.	Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO}		100	nA dc
4.	Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		0.5	V dc
5.	Base to emitter voltage (saturated)	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)3}$		1.2	V dc
6.	Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE3}	40	140	
7.	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE1}	35		
8.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	ΔI_{CBO1}	100% of initial value or 25 nA dc whichever is greater.		
9.	Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_B = 500 mA dc; pulsed (see 4.5.1)	Δh_{FE3}	±25% change from initial value.		
10.	Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$\Delta V_{CE(sat)3}$	±50 mV dc change from previous measured value.		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:
- a. Subgroup 3, see table II herein, steps 1, 3, 4, 5, 6, and 7.
- b. Subgroup 4, see table II herein, steps 2, 3, 4, 5, 6, and 10.
- c. Subgroup 5, see table II herein, steps 2, 3, 4, 5, 7, and 10.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:
- a. Subgroup 2, see table II herein, steps 1 and 6.
- b. Subgroup 3, see table II herein, steps 2 and 9.
- c. Subgroup 6, see table II herein, steps 2 and 6.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:
- a. Subgroup 2, see table II herein, steps 1, 3, 4, 5, 6, and 7 (JANS); and 1 and 6 (JANTX and JANTXV).
- b. Subgroup 3, see table II herein, steps 1, 3, 4, 5, 6, and 7 (JANS); and 1 and 6 (JANTX and JANTXV).
- c. Subgroup 6, see table II herein, steps 2, 3, 4, 5, 6, 7, 8, and 9 (JANS); and 2 and 9 (JANTX and JANTXV).

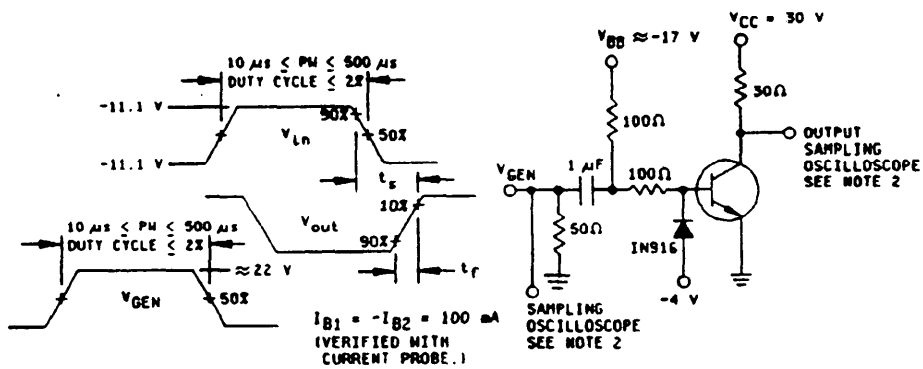
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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ .1 ns, duty cycle ≤ 2%, and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 k ohms, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 1. Test circuit and waveforms for measuring turn-on.



NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ .1 ns, duty cycle ≤ 2%, and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 k ohms, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 2. Test circuit and waveforms for measuring turn-off.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 15, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/395C	2. DOCUMENT DATE (YYMMDD) 93/11/12

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING, TYPE 2N3735, 2N3735L, and 2N3737, JANTX, JANTXV & JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial 301-731-8694 (2) AUTOVON
c. ADDRESS (Include Zip Code) NASA/Parts Projects Office (NPPO) NASA Goddard Space Flight Center Code 310.A Greenbelt, MD 20771	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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